

System Analysis and Design

IT-353

Project

DEADLINE:

PHASE 1 (MID PROJECT PROGRESS REPORT) 26/07/2026 @ 23:59

PHASE 2 (FINAL PROJECT REPORT) 9/8/2026 @ 23:59

[Total Mark is 20]

Student Details:

CRN: 50268

Name: Lana Ayman Abu Habib

ID: 240008743

Name: sondos Alansari

ID: 240051595

Name: Wajd Abdullah Alshehri

ID: 240020746

Name: shmoukh fahad alyahya

ID: 240023272

Name: Khawla Lami

ID: 240063154

Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project submission guidelines and instructions

1- Mid project Progress Report:

Students are required to submit a short mid progress report, which must include:

- a. Work completed this period
- b. One key decision has been made during this period
- c. One problem encountered
- d. Plan to do next.
- e. Each group member input/participation.
- f. A link to each group member GitHub repository

2- Version History / Commit Log Requirements:

Each group member must complete and submit project history/timestamp using a live, editable GitHub link. The repository must show commit history reflecting incremental progress. commit regularly (at minimum once per week), with clear, descriptive commit messages. A separate section in the report must be added and named "Project Artifacts" to include each student's name and a link to his/her GitHub repository (with commit history).

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also should not make a single commit containing your entire files. Both are treated as red flags regardless of the quality of the final work.

3- Written Reflection (End of Project Report):

The project report should include a Reflection section (300–500 words) as the last section (one section for each student). This section must be

written in students' own words and should address some of the following:

- a. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
- b. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
- c. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
- d. What would you do differently if you were to redo this project with more time or resources?

Important Notes

- AI use: Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. **AI must not be used to generate analysis, write code logic, or produce substantial portions of report or codebase. Any permitted AI use must be disclosed in Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.**
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit mid progress reports as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.

Project Rubric

Component	Weight	Note
Final report/deliverable	85%	Core assessment of the work itself.
Mid Progress report	10%	Incentivizes engagement across the timeline, not just at the end. Absence blocks credit for the whole project.
Version/commit history	Pass/fail	Not a grade component. Absence blocks credit for the whole project.
Reflection section	5%	Rewards demonstrated understanding and honest disclosure.

*Learning
Outcome(s):*

*Illustrate most
common analysis
and design
techniques in the
form of a group
project.*

Scenario

The Saudi Electronic University (SEU) currently organizes a wide variety of extracurricular activities, including workshops, seminars, competitions, volunteering opportunities, social events, career development sessions, and sports activities. However, information about these activities is scattered across emails, social media platforms, and announcements, making it difficult for students to discover upcoming events and register for them. Activity organizers also face challenges in managing registrations, tracking attendance, monitoring participation limits, and preparing reports.

To address these issues, SEU plans to develop a Student Activities Management System (SAMS) that will provide a centralized platform for managing all student activities. The proposed system should allow students to browse available activities by category, campus, or date, view detailed information about each activity, reserve a seat, cancel a reservation before the registration deadline, receive notifications and reminders, and record their attendance by scanning a QR code or using another electronic attendance method at the event. Students should also be able to view their personal activity history throughout their studies, including the number of activities attended, participation hours, certificates earned, and achievements. Upon graduation, the system should automatically generate a comprehensive Student Activity Report (Activity Transcript) summarizing all extracurricular activities completed during the student's academic journey.

Activity organizers should be able to create, edit, publish, and cancel activities, define participant limits, registration periods, locations (physical or virtual), attendance methods, and eligibility requirements. Organizers should also manage participant lists, approve registrations when required, record attendance, issue participation certificates, and generate reports on attendance rates and student engagement.

The Student Affairs Department should oversee the entire system by managing activity categories, approving activities submitted by organizers, monitoring participation statistics across all university branches, generating institutional reports, and ensuring that each student's activity transcript is accurate and complete before graduation.

The system should also support notifications for registration confirmations, activity reminders, cancellations, and waitlist updates. If an activity reaches its maximum capacity, additional students should automatically be placed on a waiting list and promoted when vacancies become available.

The university expects the new system to improve student engagement, simplify activity management, eliminate manual attendance tracking, and provide accurate records of students' extracurricular involvement throughout their university studies.

Based on the above scenario, complete the following tasks.

Task 1

Develop a Gantt Chart for the project, including major development activities, durations, and dependencies. [1.5 marks]

WBS ID	Level	Task Name	Start Date	End Date	Duration	Predecessor	Responsible
1.0	1	Planning and Requirements	18/07/2026	24/07/2026	7 days	—	—
1.1	2	Project Kickoff and Scenario Review	18/07/2026	19/07/2026	2 days	—	All Members
1.2	2	Define Scope, Objectives, and Stakeholders	20/07/2026	20/07/2026	1 day	1.1	Lana
1.3	2	Identify and Organize System Requirements	21/07/2026	22/07/2026	2 days	1.2	Sondos
1.4	2	Distribute Roles and Organize Project Files	23/07/2026	23/07/2026	1 day	1.3	Lana; Khawla
1.5	2	Prepare Initial Project Schedule	24/07/2026	24/07/2026	1 day	1.4	Lana
2.0	1	Phase 1 Progress Report	24/07/2026	26/07/2026	3 days	—	—

2.1	2	Draft the Mid-Project Progress Report	24/07/2026	26/07/2026	3 days	1.4	Khawla; Lana; All Members
3.0	1	Analysis and Design Deliverables	27/07/2026	05/08/2026	10 days	—	—
3.1	2	Evolutionary Prototyping and Three Low-Fidelity Screens	27/07/2026	29/07/2026	3 days	2.1	Shmoukh
3.2	2	Context Diagram and Level-0 DFD	27/07/2026	29/07/2026	3 days	2.1	Wajd
3.3	2	Three Level-1 DFDs	30/07/2026	31/07/2026	2 days	3.2	Wajd
3.4	2	Use Case Model and Three Documented Use Cases	01/08/2026	02/08/2026	2 days	3.3	Sondos
3.5	2	Two UML Activity Diagrams	03/08/2026	04/08/2026	2 days	3.4	Shmoukh
3.6	2	UML Class Diagram	03/08/2026	03/08/2026	1 day	3.4	Khawla
3.7	2	Relational Database Design and 3NF	04/08/2026	05/08/2026	2 days	3.6	Khawla
4.0	1	Final Review and Submission	06/08/2026	09/08/2026	4 days	—	—

4.1	2	Integrate All Report Sections and Diagrams	06/08/2026	06/08/2026	1 day	3.1, 3.3, 3.4, 3.5, 3.7	Lana; All Members
4.2	2	Final Consistency Review, Reflections, and Formatting	07/08/2026	07/08/2026	1 day	4.1	All Members
4.3	2	Final Project Submission — 09/08/2026 at 23:59	09/08/2026	09/08/2026	0 days	4.2	Lana

Outline Level	ID	Name	Start	Finish	Duration	Predecessors	Resource Names
1	1	Planning and Requirements	18/07/2026	24/07/2026	7 day		
2	2	Project Kickoff and Scenario Review	18/07/2026	19/07/2026	2 day		All Members
2	3	Define Scope, Objectives, and Stakeholders	20/07/2026	20/07/2026	1 day	2FS	Lana
2	4	Identify and Organize System Requirements	21/07/2026	22/07/2026	2 day	3FS	Sondos
2	5	Distribute Roles and Organize Project Files	23/07/2026	23/07/2026	1 day	4FS	Lana; Khawla
2	6	Prepare Initial Project Schedule	24/07/2026	24/07/2026	1 day	5FS	Lana
1	7	Phase 1 Progress Report	24/07/2026	26/07/2026	3 day		
2	8	Draft the Mid-Project Progress Report	24/07/2026	26/07/2026	3 day	5FS	Khawla; Lana; All Members
1	11	Analysis and Design Deliverables	27/07/2026	05/08/2026	10 day		
2	12	Evolutionary Prototyping and Three Low-Fidelity Screens	27/07/2026	29/07/2026	3 day	8FS	Shmoukh

2	13	Context Diagram and Level-0 DFD	27/07/2026	29/07/2026	3 day	8FS	Wajd
2	14	Three Level-1 DFDs	30/07/2026	31/07/2026	2 day	13FS	Wajd
2	15	Use Case Model and Three Documented Use Cases	01/08/2026	02/08/2026	2 day	14FS	Sondos
2	16	Two UML Activity Diagrams	03/08/2026	04/08/2026	2 day	15FS	Shmoukh
2	17	UML Class Diagram	03/08/2026	03/08/2026	1 day	15FS	Khawla
2	18	Relational Database Design and 3NF	04/08/2026	05/08/2026	2 day	17FS	Khawla
1	19	Final Review and Submission	06/08/2026	09/08/2026	4 day		
2	20	Integrate All Report Sections and Diagrams	06/08/2026	06/08/2026	1 day	12FS,14FS,15FS,16FS,18FS	Lana; All Members
2	21	Final Consistency Review, Reflections, and Formatting	07/08/2026	07/08/2026	1 day	20FS	All Members
2	22	Final Project Submission — 09/08/2026 at 23:59	09/08/2026	09/08/2026	0 day	21FS	Lana

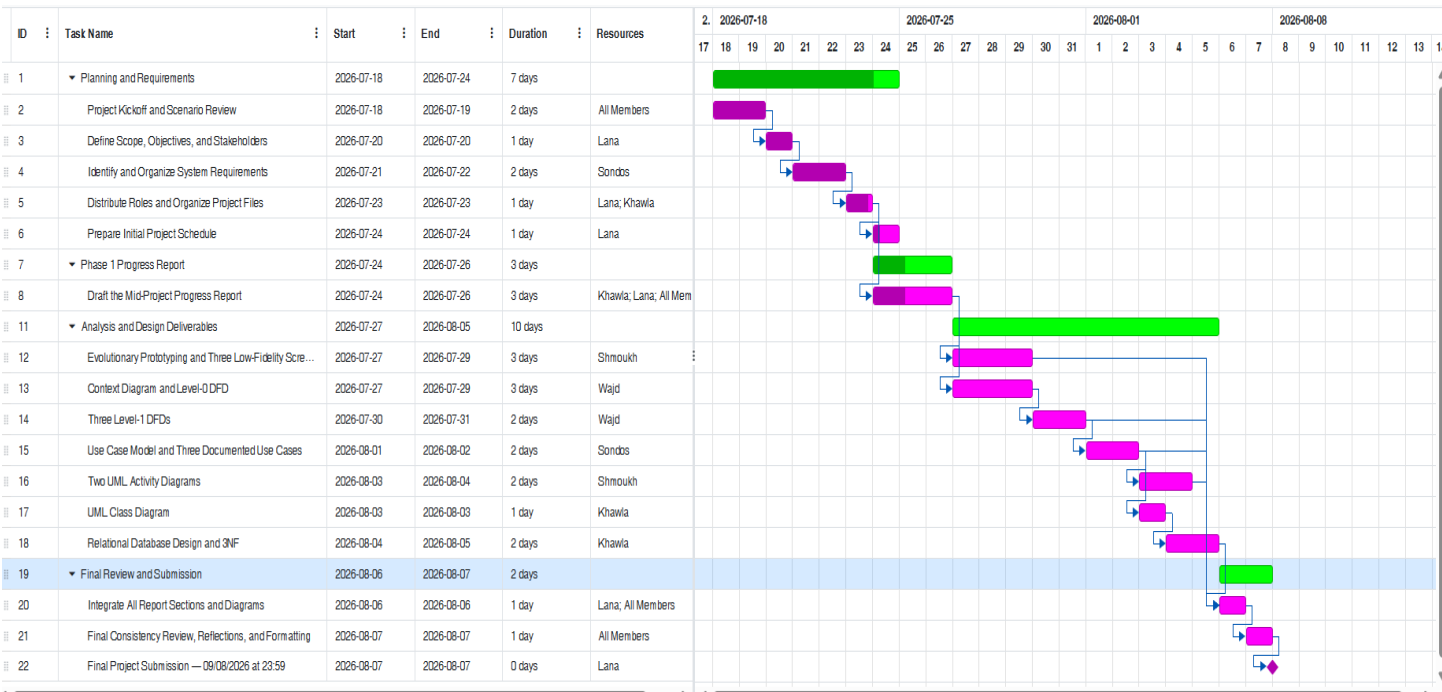


Figure 1 Gant Chart

Task 2

Select ONE prototyping approach (Evolutionary Prototyping or Throwaway Prototyping). Develop low-fidelity prototypes for 3 main screens of the system and justify why your chosen approach is more appropriate for this project. [2 marks]

2.1 Selected Approach: Evolutionary Prototyping

For the SAMS project, we selected **Evolutionary Prototyping**. In this approach, we start by developing the important parts of the system, then users test the prototype and give their feedback. After that, we keep improving it until the prototype become the final working system. The prototype is not deleted like Throwaway Prototyping.

This approach is more suitable for SAMS because:

- SAMS is a new system, so some requirements may not be clear from the beginning. When students and staff test the prototype, they may remember new requirements or ask to change some parts.
- The system has different users, such as students, organizers, and Student Affairs staff. Each user may have different needs, so their feedback will help us improve the screens and functions.
- Some functions need more testing, like the waiting list, QR attendance, eligibility rules, and approving activities. It is better to build these parts early and improve them step by step.
- The prototype can show if the screens are crowded, confusing, or difficult to use. For example, users can test the activity search, seat reservation, and activity management screens before completing the whole system.
- The work will not be wasted because every improved version can be used in the final SAMS system. This is better for our project because the development time is limited.

For these reasons, Evolutionary Prototyping can help us understand the requirements better and build a system that is more useful for the actual users.

2.2 Prototype 1 — Student: Browse Activities

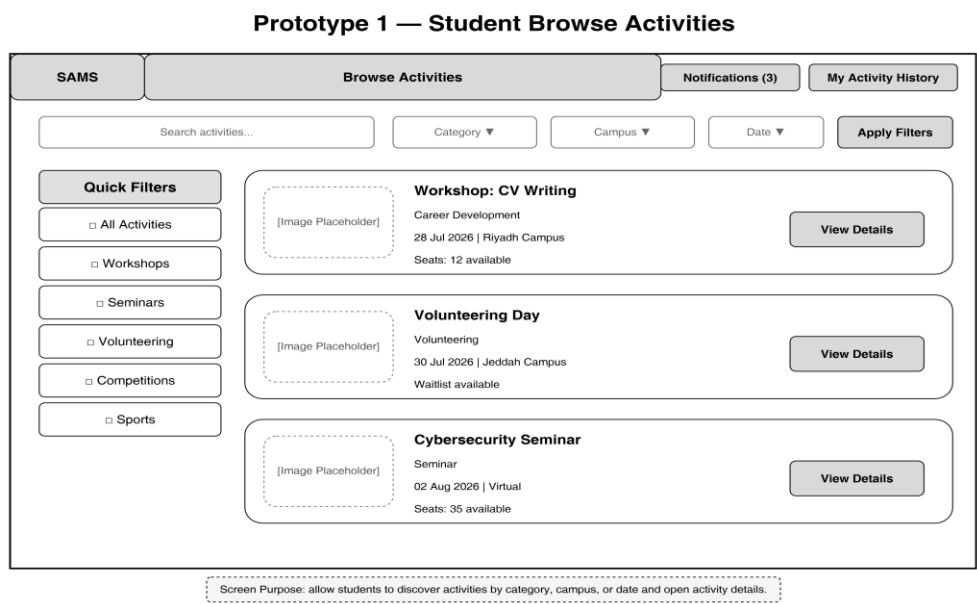


Figure 2 Prototype 1 — Student: Browse Activities

2.3 Prototype 2 — Activity Details and Reservation

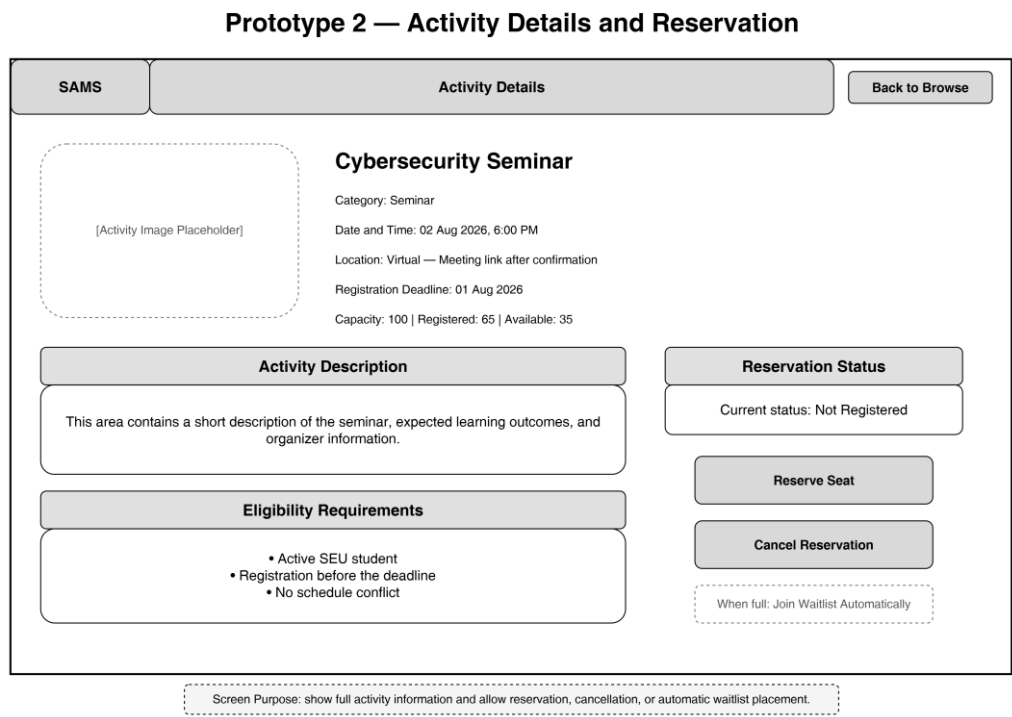


Figure 3 Prototype 2 — Activity Details and Reservation

2.4 Prototype 3 — Organizer Activity Management

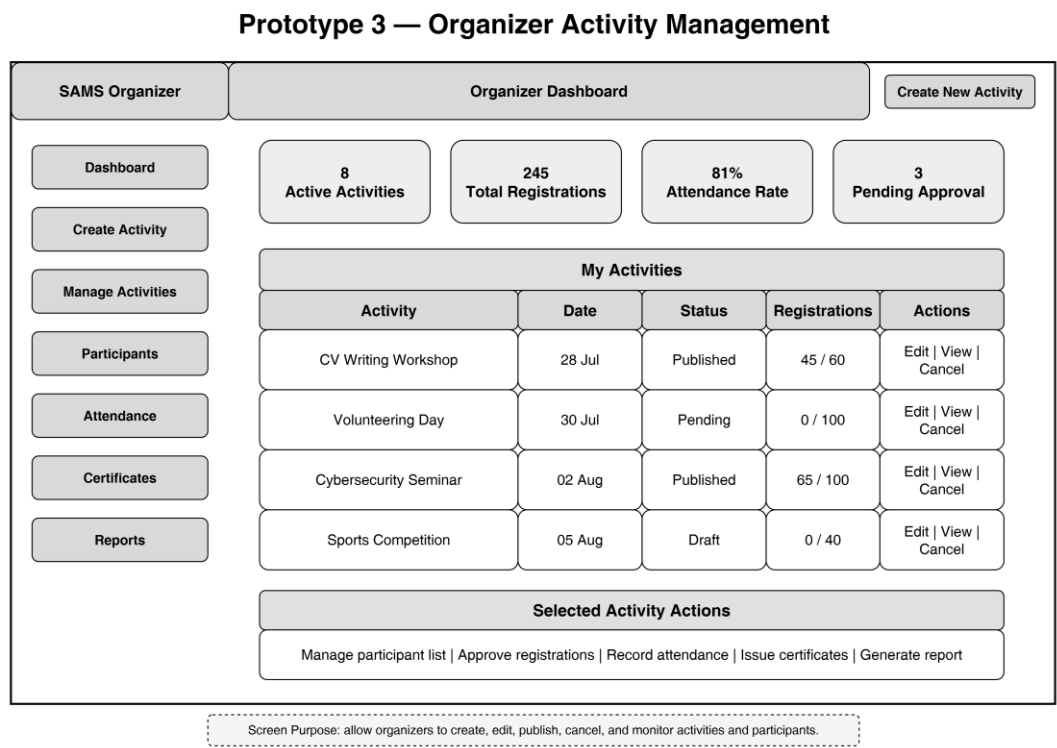


Figure 4 Prototype 3 — Organizer Activity Management